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A model implementation of a scalable data store for scientific computing with DataLad

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Domain-agnostic representation in DataLad RIA stores enables separation of scientific workflows and maintenance/data curation by non-scientific personnel

1 Prepare analysis on office desktop

Analysis A — my_analysisA/ @master_branch
- code/
- results/ (from result_branch)
  - important_table.csv
  - image_overlay.tif
  - ...
- README.md
- ...
- BIDS — inputs/
- environments/
  - labworkflows/
    - stdtools_debian_9.simg
    - ...

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DataLad can obtain required subdataset content on demand. Only content elements actually required for an analysis are present. Directory structure is expanded recursively as needed.

Images of containerized computational environments are tracked in version control just like any other data file. Actual storage can be local or in cloud.

put into store

2
- `datalad create-sibling-ria` to ria+ssh://store.domain.my
- `datalad push` current state

3
- `datalad clone --reckless ephemeral` from ria+file:///store#~analysisA
- `git checkout -b result_branch`

get from store

update in store

4 Run analysis on compute node

Analysis A — analysisA/ @result_branch
- code/
- results/
  - important_table.csv
  - image_overlay.tif
  - ...
- README.md
- ...
- BIDS — inputs/
  - dataset_description.json
  - sub-001/
  - ...
  - sourcedata/
- RAW — my_precious_dicoms/
- .git/
  - annex/ -> /store/2e5/24934-a09e-11e9-8503-f0d5bf7b5561/

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Any input data is referenced via the dataset that contains it. Dataset state provides unambiguous version specification for any data dependency.

5
- `datalad push` the results
- `datalad remove` local clone

6
- `datalad update` to fetch results
- `git merge` to keep results

update from store

update in store

7
- `datalad push` to store
- `datalad remove` local clone

SSH

same file system

Adminstrator

- enables additional adminstrative tasks
- git/git-annex based assessment of data structure
- can answer questions like:
  - what is actually used? by whom?
  - what can be archived?
  - who created/changed what when/last?
  - what does a project depend on?
- enables maintenance tasks:
  - garbage collection (clean unreferenced git/git-annex objects)
  - find data duplication based on content hashes
  - maintain (group) permissions

admin

Writing paper on Laptop@Home

Paper — my_high_impact_paper/
- code/
- plots/
- tex/
- submission.pdf
- ...
- Analysis A — analysisA/
  - code/
  - results/
  - ...
  - BIDS — inputs/
- Analysis B — analysisB/
  - code/
  - results/
  - ...

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8
- `datalad clone` from ria+http:///store.domain.my#~analysisA

get from store

Mirrored store @webserver

store/
- 2e5/
  - 24934-a09e-11e9-8503-f0d5bf7b5561/

use multiple (synchronized) stores to physically separate different access methods/permissions

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RIA Datastore layout

store/

Analysis A — 2e5/24934-a09e-11e9-8503-f0d5bf7b5561/
- branches
- config
- description
- HEAD
- refs/
  - ...
- objects/
  - ...
- annex/
  - ...
  - objects/
    - ...
    - 5X/
      - Xg/
        - MD5E-s4--3cd7a0db76ff9dca48979e24c39b408c.nii/
          - MD5E-s4--3cd7a0db76ff9dca48979e24c39b408c.nii
- archives/
  - archive.7z

Analysis B — 62e/9a606-abc9-11ea-bd40-3b17196da88a/

BIDS — 64d/ddedc-abc9-11ea-bd40-3b17196da88a/

RAW — 67e/a7e00-abc9-11ea-bd40-3b17196da88a/

Paper — fea/f8808-abc9-11ea-bd40-3b17196da88a/

Unused — b2e/409f8-aa3a-11ea-a930-3d4ce1d1b284/

...

alias/
- analysisA -> ../2e5/24934-a09e-11e9-8503-f0d5bf7b5561/
- ...

The dataset ID determines a dataset's location in the data store, avoiding (careless) duplicates.

A bare Git repository holds a clone of the dataset.

The git-annex object tree contains (large) data.

Optional 7z archive with (additional) git-annex objects. This allows to represent the entire dataset with about 20 inodes.

subdataset references

git-annex objects can be moved into archives/archive.7z to lower inode consumption

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recently used dataset; regard all (data) dependencies as in use

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unreferenced dataset; wasn't touched in ages
can be moved out of store to longterm archival storage

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provide convenience aliases

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Researcher

- access via local file system, SSH, HTTP depending on location
- set up project/analysis datasets w/ data dependencies
- fetch data updates
- (re-)run analysis
- push (analysis) results
- remove local copies/clones

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Data curator

- create DataLad datasets from available raw data
- create/update (BIDS) standard representation (as derived datasets)
- set (group) permissions for data access in accordance to legal requirements
- archive publication w/ all data dependencies
- remove local copies/clones

curator

DataLad

Documentation:
http://handbook.datalad.org

DataLad on Github:
https://github.com/datalad/datalad

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